

Marthan Lanuzga

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Education

Catanduanes State University
BS Computer Engineering

Catanduanes, PH
2022 - Present

- **1st Place Winner** (Team Leader) – 2026 Regional ICpEP Robothon Challenge.
- **Regional Representative** — 2023 Regional ICpEP Robothon Challenge.

Projects

Pathfinder | Autonomous Tourism Agent
Full-Stack AI Developer

September 2025 – Present

- Created a greedy heuristic algorithm to approximate the **Traveling Salesperson Problem (TSP)** on edge devices, clustering destinations geographically while adhering to **$O(n)$ complexity constraints**.
- Reduced **Retrieval-Augmented Generation (RAG) pipeline latency by 90% (to ~300ms)** by architecting a semantic caching layer on ChromaDB, eliminating redundant vector computations and cutting LLM operational costs by **~90%**.
- Designed an **offline-first** distributed architecture deployed on Raspberry Pi hardware, achieving **99% system availability** and enabling query processing in zero-connectivity environments.

Med-ID | AI Drug Label Analyzer
Backend Developer

September 2025 – October 2025

- Engineered an **automated data pipeline** using **Roboflow** to clean, annotate, and augment a dataset of raw medication images, improving model robustness against real-world noise and lighting variations.
- Developed a **Convolutional Neural Network (CNN)** in **TensorFlow/Keras** to classify raw inputs.
- Architected a **hybrid computer vision pipeline** using **OpenCV** for adaptive image thresholding and **PaddleOCR** for text extraction on high-noise medical imagery.
- Trained a custom **NER model** using **spaCy** to extract pharmacological entities, cross-referencing outputs against the **OpenFDA API** to ensure 100% data compliance.
- Deployed a **semantic analysis engine** via **Google Gemini API**, utilizing prompt engineering to convert structured technical data into patient-friendly safety summaries.

- Architected an end-to-end e-commerce storefront using Next.js and Supabase for real-time inventory management, authentication, and image storage.
- Integrated the PayMongo API to handle secure local payment processing, including GCash and Maya support via specialized webhook listeners.
- Developed responsive UI using Tailwind CSS, implementing server-side logic for order validation and automated stock restocking via Next.js API routes.

Experience

DOST – Advance Science and Technology Institute
Software Development Intern

Quezon City, PH
May 2025 – August 2025

- Scalability-tested and optimized ML pipelines for large-scale computer vision projects, ensuring 100% compliance with ISO-compliant government thresholds while reducing manual feature extraction by 40%.
- Collaborated in an Agile team of 9, integrating React interfaces with ML models and delivering key features 10% ahead of schedule.
- Optimized ML pipelines by replacing arbitrary parameters with ISO-compliant thresholds, ensuring 100% alignment with validated government standards.

Sasquatch
Founder

Metro Manila, PH
December 2025 – Present

- Architected a **behavior-aware AI assistant** that **eliminated 95%** of common syntax and type errors during the development phase via local deterministic analysis, maintaining a sub-50ms execution overhead.
- Designed a deterministic-first architecture with clear separation between local analysis and a future AI escalation layer.

Skills & Interests

- **Technical:** Python, SQL, JavaScript, HTML/CSS, C++, TypeScript
- **Frameworks/Libraries:** FastAPI, Next.js, Supabase, Tailwind CSS
- **Web Development:** React.js, Vue
- **AI, ML, and Data Science:** TensorFlow, PyTorch, Keras, OpenCV, pandas
- **Tools/Databases:** Git/GitHub, PostgreSQL, Docker, RAG, ChromaDB, SQLite, MySQL
- **Interests:** AI Agent Orchestration, Autonomous Systems, Web Development, AI & Data Science